statement. The gene pool will become an *evolutionarily stable set* of genes, defined as a gene pool that cannot be invaded by any new gene. Most new genes that arise, either by mutation or reassortment or immigration, are quickly penalized by natural selection: the evolutionarily stable set is restored. Occasionally a new gene does succeed in invading the set: it succeeds in spreading through the gene pool. There is a transitional period of instability, terminating in a new evolutionarily stable set—a little bit of evolution has occurred. By analogy with the aggression strategies, a population might have more than one alternative stable point, and it might occasionally flip from one to another. Progressive evolution may be not so much a steady upward climb as a series of discrete steps from stable plateau to stable plateau.* It may look as though the population as a whole is behaving like a single self-regulating unit. But this illusion is produced by selection going on at the level of the single gene. Genes are selected on 'merit'. But merit is judged on the basis of performance against the background of the evolutionarily stable set which is the current gene pool.

By focussing on aggressive interactions between whole individuals, Maynard Smith was able to make things very clear. It is easy to think of stable ratios of hawk bodies and dove bodies, because bodies are large things which we can see. But such interactions between genes sitting in *different* bodies are only the tip of the iceberg. The vast majority of significant interactions between genes in the evolutionarily stable set—the gene pool—go on *within* individual bodies. These interactions are difficult to see, for they take place within cells, notably the cells of developing embryos. Well-integrated bodies exist because they are the product of an evolutionarily stable set of selfish genes.

But I must return to the level of interactions between whole animals which is the main subject of this book. For understanding aggression it was convenient to treat individual animals as independent selfish machines. This model breaks down when the individuals concerned are close relatives—brothers and sisters, cousins, parents and children. This is because relatives share a substantial proportion of their genes. Each selfish gene therefore has its loyalties divided between different bodies. This is explained in the next chapter.